

NumPy Assignment One

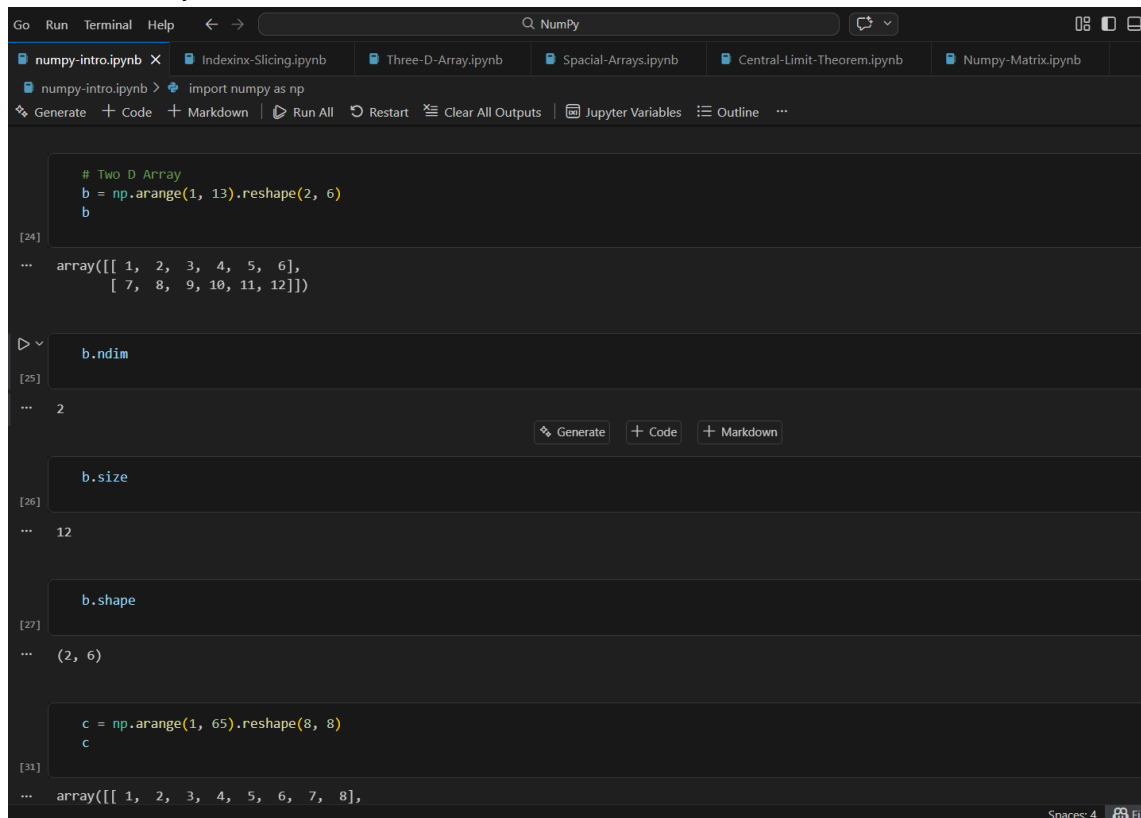
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What is NumPy? Why should we use it?

NumPy stands for numerical python. It is a library in python that is used for scientific and mathematical calculations. We should use it for statistical calculations. It has wide range of applications in deep learning and computer vision.

Write the steps to create 2D, and 3D array with output.

a. Two-D Array:



```
Go Run Terminal Help  NumPy
numpy-intro.ipynb x  Indexix-Slicing.ipynb  Three-D-Array.ipynb  Spacial-Arrays.ipynb  Central-Limit-Theorem.ipynb  Numpy-Matrix.ipynb
numpy-intro.ipynb > import numpy as np
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline ...

# Two D Array
b = np.arange(1, 13).reshape(2, 6)
b

[24]
... array([[ 1,  2,  3,  4,  5,  6],
          [ 7,  8,  9, 10, 11, 12]])

b.ndim

[25]
... 2

b.size

[26]
... 12

b.shape

[27]
... (2, 6)

c = np.arange(1, 65).reshape(8, 8)
c

[31]
... array([[ 1,  2,  3,  4,  5,  6,  7,  8],
```

b. Three-D Array:

```
> import numpy as np
1]
>
2] a = np.arange(25, 201, 4).reshape(2, 11, 2)
a
array([[[ 25, 29],
        [ 33, 37],
        [ 41, 45],
        [ 49, 53],
        [ 57, 61],
        [ 65, 69],
        [ 73, 77],
        [ 81, 85],
        [ 89, 93],
        [ 97, 101],
        [105, 109]],
       [[113, 117],
        [121, 125],
        [129, 133],
        [137, 141],
        [145, 149],
        [153, 157],
        [161, 165],
        [169, 173],
        [177, 181],
        [185, 189],
        [193, 197]]])
```

How can you identify the datatype of a given NumPy array?

how do I change the data type of an array?

Every NumPy array carries its datatype in the dtype attribute:

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(arr.dtype)
```

NumPy provides a clean way to convert types using astype():

```
arr = np.array([1, 2, 3, 4])
arr_float = arr.astype(float)
print(arr_float)
print(arr_float.dtype)
```

Create an array using in numpy. Then convert numeric array to a categorical (text) array.

```
import numpy as np
arr = np.array([0, 1, 2, 1, 0])
labels = np.array(["Low", "High", "Medium"])
```

```
category = labels[arr]  
print(category.dtype)
```